

CELESTIAL OBJECTS

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THE UNIVERSE CONSISTS of energy, space, and matter. Some of the matter drifts through space as single atoms or simple gas molecules. Other matter clumps into islands of material, from dust motes to giant suns, or implodes to form black holes. Gravity binds all of these objects into the great clouds and disks of material known as galaxies. Galaxies in turn fall into clusters and finally form the biggest celestial objects of all—superclusters.



DARK NEBULA
Also known as the Boogeyman Nebula, LDN 1622 is an extensive cloud of dust and gas. It lies about 500 light-years away.

GAS, DUST, AND PARTICLES

Much of the ordinary matter of the universe exists as a thin and tenuous gas within and around galaxies and as an even thinner gas between galaxies. The gas is made mainly of hydrogen and helium atoms, but some clouds inside galaxies contain atoms of heavier chemical elements and simple molecules. Mixed in with the galactic gas clouds is dust—tiny solid particles of carbon or substances such as silicates (compounds of silicon and oxygen). Within galaxies, the gas and dust make up what is called the interstellar medium. Visible clumps of this medium, many of them the sites of star formation, are called nebulae. Some, called emission nebulae, produce a brilliant glow as their constituent atoms absorb radiant energy from stars and reradiate it as light. In contrast, dark nebulae are visible only as smudges that block out starlight. Particles of matter also exist in space in the form of cosmic rays—highly energetic subatomic particles traveling at high speed through the cosmos.



STAR-FORMING NEBULA
The Carina Nebula, a giant cloud of gas, is a prominent feature of the sky in the Southern Hemisphere and is visible to the naked eye. Different colors in this image represent temperature variations in the gas.

GLOWING GAS
This ocean of glowing gas is an active region of star formation in the Omega Nebula, an emission nebula. Clouds of gas and dust may give birth to stars and planets, but they are also cast off by dying stars, eventually to be recycled into the next stellar generation.